

AAE 50700: Principles of Dynamics
Problem Set 4 Report

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(3.1) Constraints, Virtual Work, and D'Alembert's Principle

Consider a rigid bus B with a rigid solar array attached via a hinge. Fig. 1 shows a top-down view of the satellite. A rigid, thin solar-array paddle A is connected to B via a friction-less hinge H on B . Both Problem Set 3 and Problem Set 4 will focus on the planar motion of the satellite in the inertial frame $\mathcal{N}''\{\hat{n}_1, \hat{n}_2\}$.

Relevant quantities and parameters are defined as follows:

- **Bus:** C_B , m_B , and J_{C_B} denote the bus center of mass (CoM), mass, and moment of inertia about C_B , respectively. The bus is a cube of side length $2l$ and of uniform density.
- **Array:** C_A , m_A , and J_{C_A} denote the array CoM, mass, and moment of inertia about C_A respectively. The $H - C_A$ distance is d .

The positions of C_B and C_A measured from the origin are given by:

$$\mathbf{r}_{C_B}(t) = x_{C_B}(t)\hat{n}_1 + y_{C_B}(t)\hat{n}_2 \quad (1)$$

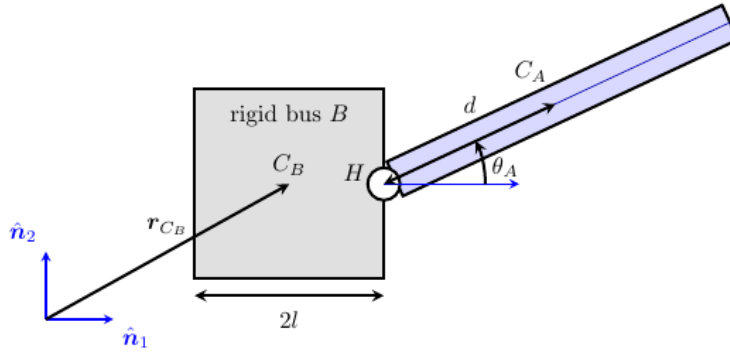


Figure 1: Top-down view of satellite bus B and solar array A , which are both assumed rigid. Both Problem Set 3 and Problem Set 4 consider this system.

(4.1) Lagrange's Equations

(a): Given

Starting from Lagrangian form of D'Alembert's principle.

Find:

Derive the Lagrange's equations for non-conservative systems with minimal coordinates $\mathbf{q}$.

Analysis/Conclusions:

The Lagrangian form of D'Alembert's principle is written as:

$$\sum_{i=1}^N (\mathbf{F}_i - m_i \ddot{\mathbf{r}}_i) \cdot \delta \mathbf{r}_i = 0 \quad (2)$$

where $\mathbf{F}_i$ are the applied forces, since the constraint forces do no virtual work. Suppose the positions can be expressed via generalized coordinates $\mathbf{q}$ as $\mathbf{r}_i = \mathbf{r}_i(\mathbf{q}, t)$. Since virtual displacements occur without the passage of time ($\delta t = 0$), the virtual displacement is:

$$\delta \mathbf{r}_i = \sum_{j=1}^n \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j \quad (3)$$

Substituting this into the virtual work of the applied forces and exchanging the order of summation:

$$\sum_{i=1}^N \mathbf{F}_i \cdot \delta \mathbf{r}_i = \sum_{i=1}^N \mathbf{F}_i \cdot \left(\sum_{j=1}^n \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j \right) = \sum_{j=1}^n \left(\sum_{i=1}^N \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right) \delta q_j = \sum_{j=1}^n Q_j \delta q_j \quad (4)$$

where Q_j is the generalized force, given below in Eq. (8).

Similarly, the inertial term becomes:

$$\sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \delta \mathbf{r}_i = \sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \left(\sum_{j=1}^n \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j \right) = \sum_{j=1}^n \left(\sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right) \delta q_j \quad (5)$$

Combining both terms, D'Alembert's principle becomes:

$$\sum_{j=1}^n \left(Q_j - \sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right) \delta q_j = 0 \quad (6)$$

Since $\mathbf{q}$ is a minimal coordinate set, the δq_j are independent from each other, so the sums can be written independently.

Which gives us the Lagrangian form of D'Alembert's principle for minimal coordinates is written as:

$$\sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = Q_j \quad \forall j \quad (7)$$

With mass m_i , position $\mathbf{r}_i$, and N representing the number of particles in a model.

The generalized force term Q_j can be represented by:

$$Q_j = \sum_{i=1}^N \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \quad (8)$$

The core of Lagrange's idea was to use the partial derivative of kinetic energy T with respect to $(q_j, \dot{q}_j)$ with the above D'Alembert's equations. Recall the general expression for kinetic energy is:

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i \quad (9)$$

Then we find the two derivatives as:

$$\begin{aligned} \frac{\partial T}{\partial q_j} &= \frac{\partial}{\partial q_j} \left(\frac{1}{2} \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i \right) = \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial q_j} \\ \frac{\partial T}{\partial \dot{q}_j} &= \frac{\partial}{\partial \dot{q}_j} \left(\frac{1}{2} \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i \right) = \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} \end{aligned} \quad (10)$$

Then, since $\mathbf{r}_i(\mathbf{q}, t)$ is not a function of the derivatives of q_j , then the cancellation of dots identity applies where:

$$\frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{r}_i}{\partial q_j} \quad (11)$$

Using this identity, Eq. (8) can be rewritten as:

$$Q_j = \sum_{i=1}^N \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = \sum_{i=1}^N \mathbf{F}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} \quad (12)$$

Then we can take the time derivative found in the second equation in Eq. (10).

Note the application of the cancellation of dots identity on line three:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) &= \frac{d}{dt} \left(\sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} \right) \\
&= \sum_{i=1}^N m_i \frac{d\dot{\mathbf{r}}_i}{dt} \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \frac{d}{dt} \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} \\
&= \sum_{i=1}^N m_i \frac{d\dot{\mathbf{r}}_i}{dt} \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \frac{d}{dt} \frac{\partial \mathbf{r}_i}{\partial q_j} \\
&= \sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} + \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial q_j}
\end{aligned} \tag{13}$$

Then we notice two things: first, since Eq. (7) gives $\sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = Q_j$, we see that the left half of Eq. (13), is equal to the term that came from cancellation of dots on Q_j in Eq. (12). Second, we see that the right half of Eq. (13) is equal to the first equation from the derivatives in Eq. (10). In other words:

$$Q_j = \sum_{i=1}^N m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}, \quad \frac{\partial T}{\partial q_j} = \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial q_j} \tag{14}$$

Which then means the last line of Eq. (13) substitutes to:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) &= Q_j + \frac{\partial T}{\partial q_j} \\
Q_j &= \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j}
\end{aligned} \tag{15}$$

Which is the fundamental form of Lagrange's equations for non-conservative systems with minimal coordinates $\mathbf{q}$.

Then, to get the generalized Lagrange equation, we introduce what is known as the Lagrangian function, where L is the Lagrangian, T is still kinetic energy, and V is potential energy:

$$L(\mathbf{q}, \dot{\mathbf{q}}, t) = T(\mathbf{q}, \dot{\mathbf{q}}, t) - V(\mathbf{q}, t) \tag{16}$$

Which then rewrites to $T = L + V$. When substituted into the partial derivatives in Eq. (15) we get:

$$\begin{aligned}
\frac{\partial T}{\partial q_j} &= \frac{\partial L + V}{\partial q_j} = \frac{\partial L}{\partial q_j} + \frac{\partial V}{\partial q_j} \\
\frac{\partial T}{\partial \dot{q}_j} &= \frac{\partial L + V}{\partial \dot{q}_j} = \frac{\partial L}{\partial \dot{q}_j} + \frac{\partial V}{\partial \dot{q}_j} = \frac{\partial L}{\partial \dot{q}_j}
\end{aligned} \tag{17}$$

Which happens since L is a function of $\mathbf{q}$ and $\dot{\mathbf{q}}$ while V is just a function of $\mathbf{q}$,

Then when substituting back into the fundamental form, we get:

$$\begin{aligned} Q_j &= \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \\ &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} - \frac{\partial V}{\partial q_j} \end{aligned} \quad (18)$$

The generalized force Q_j makes no assumption about the applied forces. If the applied forces consist of conservative and non-conservative forces, i.e. $\mathbf{F}_i = \mathbf{F}_{\text{nc},i} - \frac{\partial V}{\partial \mathbf{r}_i}$, then substituting into Eq. (8), the generalized forces become:

$$\begin{aligned} Q_j &= \sum_{i=1}^N \left(\mathbf{F}_{\text{nc},i} - \frac{\partial V}{\partial \mathbf{r}_i} \right) \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \\ &= \sum_{i=1}^N \mathbf{F}_{\text{nc},i} \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} - \sum_{i=1}^N \frac{\partial V}{\partial \mathbf{r}_i} \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \\ &= Q_{\text{nc},j} - \frac{\partial V}{\partial q_j} \end{aligned} \quad (19)$$

where $Q_{\text{nc},j} = \sum_{i=1}^N \mathbf{F}_{\text{nc},i} \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}$, and the last step uses the chain rule, since V depends on $\mathbf{q}$ only through the positions $\mathbf{r}_i$.

Substituting this into the fundamental form with the L derivatives found above:

$$\begin{aligned} Q_{\text{nc},j} - \frac{\partial V}{\partial q_j} &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} - \frac{\partial V}{\partial q_j} \\ Q_{\text{nc},j} &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} \end{aligned} \quad (20)$$

Which gives the Lagrange's equations for non-conservative systems with minimal coordinates $\mathbf{q}$ as:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = Q_{\text{nc},j} \quad \forall j \quad (21)$$

Because this was derived from the minimal coordinate version of the Lagrangian form of D'Alembert's principle, and it incorporates non-conservative generalized force.

(b): Given

Consider array A attached to the bus B that is moving in the same manner as in Eq. (22) (translational motion only).

$$\mathbf{r}_{C_B}(t) = X_B \cos \Omega t \hat{\mathbf{n}}_1, \quad \mathbf{r}_H(t) = \mathbf{r}_{C_B}(t) + l \hat{\mathbf{n}}_1 \quad (22)$$

The array has a torsional spring at H with potential:

$$V_H(\theta_A) = \frac{k_H}{2} (\theta_A - \theta_0)^2 \quad (23)$$

(b.1): Find

Derive the expression of $\mathbf{r}_{C_A}$ in terms of $\mathbf{r}_H$, d , θ_A , $\hat{\mathbf{n}}_1$, $\hat{\mathbf{n}}_2$.

Analysis/Conclusions:

Recall from problem set 3, we can write the unit vector pointing from the hinge H to the array CoM C_A using:

$$\hat{\mathbf{e}}_A = \cos \theta_A \hat{\mathbf{n}}_1 + \sin \theta_A \hat{\mathbf{n}}_2 \quad (24)$$

Then we can define the position of $\mathbf{r}_{C_A}$ as a sum of the position of the hinge $\mathbf{r}_H$ and the vector $\hat{\mathbf{e}}_A$ scaled by d . This is because $\mathbf{r}_H$ offsets the coordinates to the hinge position, $\hat{\mathbf{e}}_A$ points towards the array, and d scales that unit vector to the distance of the array CoM from the hinge. This is written as:

$$\mathbf{r}_{C_A} = \mathbf{r}_H + d \hat{\mathbf{e}}_A \quad (25)$$

Then substituting this in with the given definition of $\hat{\mathbf{e}}_A$, we get the equation for $\mathbf{r}_{C_A}$:

$$\mathbf{r}_{C_A} = \mathbf{r}_H + d (\cos \theta_A \hat{\mathbf{n}}_1 + \sin \theta_A \hat{\mathbf{n}}_2) \quad (26)$$

(b.2): Find

Calculate the kinetic energy:

$$T_A = \frac{m_A}{2} \dot{\mathbf{r}}_{C_A} \cdot \dot{\mathbf{r}}_{C_A} + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \quad (27)$$

and form $L_A(\theta_A, \dot{\theta}_A, t) = T_A - V_H$.

Analysis/Conclusions:

To find the kinetic energy from the given equations, we must first find the first derivative of $\mathbf{r}_{C_A}$:

$$\begin{aligned} \dot{\mathbf{r}}_{C_A} &= \frac{d}{dt} (\mathbf{r}_H + d(\cos \theta_A \hat{\mathbf{n}}_1 + \sin \theta_A \hat{\mathbf{n}}_2)) \\ &= \frac{d\mathbf{r}_H}{dt} + d \left(\frac{d}{d\theta_A} \cos \theta_A \frac{d\theta_A}{dt} \hat{\mathbf{n}}_1 + \frac{d}{d\theta_A} \sin \theta_A \frac{d\theta_A}{dt} \hat{\mathbf{n}}_2 \right) \\ &= \dot{\mathbf{r}}_H - d \sin \theta_A \dot{\theta}_A \hat{\mathbf{n}}_1 + d \cos \theta_A \dot{\theta}_A \hat{\mathbf{n}}_2 \end{aligned} \quad (28)$$

And the first derivative of $\mathbf{r}_H$:

$$\begin{aligned} \dot{\mathbf{r}}_H &= \frac{d}{dt} (\mathbf{r}_{C_B} + l \hat{\mathbf{n}}_1) \\ &= \frac{d}{dt} (X_B \cos \Omega t \hat{\mathbf{n}}_1 + l \hat{\mathbf{n}}_1) \\ &= X_B \frac{d}{dt} \cos \Omega t \hat{\mathbf{n}}_1 + \frac{d}{dt} l \hat{\mathbf{n}}_1 \\ &= -X_B \Omega \sin \Omega t \hat{\mathbf{n}}_1 \end{aligned} \quad (29)$$

Then compute the dot product in Eq. (27) by substituting in the two derivatives:

$$\begin{aligned} \dot{\mathbf{r}}_{C_A} \cdot \dot{\mathbf{r}}_{C_A} &= \left(\dot{\mathbf{r}}_H - d \sin \theta_A \dot{\theta}_A \hat{\mathbf{n}}_1 + d \cos \theta_A \dot{\theta}_A \hat{\mathbf{n}}_2 \right) \cdot \left(\dot{\mathbf{r}}_H - d \sin \theta_A \dot{\theta}_A \hat{\mathbf{n}}_1 + d \cos \theta_A \dot{\theta}_A \hat{\mathbf{n}}_2 \right) \\ &= \left(-X_B \Omega \sin \Omega t - d \sin \theta_A \dot{\theta}_A \right)^2 + \left(d \cos \theta_A \dot{\theta}_A \right)^2 \\ &= X_B^2 \Omega^2 \sin^2 \Omega t + 2X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A + d^2 \sin^2 \theta_A \dot{\theta}_A^2 + d^2 \cos^2 \theta_A \dot{\theta}_A^2 \\ &= X_B^2 \Omega^2 \sin^2 \Omega t + 2X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A + (\sin^2 \theta_A + \cos^2 \theta_A) d^2 \dot{\theta}_A^2 \\ &= X_B^2 \Omega^2 \sin^2 \Omega t + 2X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A + d^2 \dot{\theta}_A^2 \end{aligned} \quad (30)$$

Then we can find the kinetic energy T_A :

$$\begin{aligned} T_A &= \frac{m_A}{2} \dot{\mathbf{r}}_{C_A} \cdot \dot{\mathbf{r}}_{C_A} + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \\ &= \frac{m_A}{2} \left(X_B^2 \Omega^2 \sin^2 \Omega t + 2X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A + d^2 \dot{\theta}_A^2 \right) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \\ &= \frac{m_A}{2} \left(X_B^2 \Omega^2 \sin^2 \Omega t + 2X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A \right) + \frac{m_A d^2 + J_{C_A}}{2} \dot{\theta}_A^2 \end{aligned} \quad (31)$$

So, the scalar kinetic energy equation can be written as:

$$T_A = \frac{m_A}{2} X_B^2 \Omega^2 \sin^2 \Omega t + m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A + \frac{m_A d^2 + J_{C_A}}{2} \dot{\theta}_A^2 \quad (32)$$

Then, finding the Lagrangian $L_A(\theta_A, \dot{\theta}_A, t)$ is as simple as substituting Eq. (32) and Eq. (23) into the following:

$$\begin{aligned} L_A &= T_A - V_H \\ &= \frac{m_A}{2} X_B^2 \Omega^2 \sin^2 \Omega t + m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A \\ &\quad + \frac{m_A d^2 + J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \end{aligned} \quad (33)$$

(b.3): Find

Verify the cancellation-of-dots identity $\frac{\partial \dot{\mathbf{r}}_{C_A}}{\partial \dot{\theta}_A} = \frac{\partial \mathbf{r}_{C_A}}{\partial \theta_A}$.

Analysis/Conclusions:

To verify the identity, first we compute $\frac{\partial \mathbf{r}_{C_A}}{\partial \theta_A}$ using the equation found in part (b.1). Recall that $\mathbf{r}_H$ is not a function of θ_A :

$$\begin{aligned} \frac{\partial \mathbf{r}_{C_A}}{\partial \theta_A} &= \frac{\partial}{\partial \theta_A} (\mathbf{r}_H + d(\cos \theta_A \hat{\mathbf{n}}_1 + \sin \theta_A \hat{\mathbf{n}}_2)) \\ &= \frac{\partial \mathbf{r}_H}{\partial \theta_A} + d \left(\frac{\partial}{\partial \theta_A} \cos \theta_A \hat{\mathbf{n}}_1 + \frac{\partial}{\partial \theta_A} \sin \theta_A \hat{\mathbf{n}}_2 \right) \\ &= d(-\sin \theta_A \hat{\mathbf{n}}_1 + \cos \theta_A \hat{\mathbf{n}}_2) \end{aligned} \quad (34)$$

Then do the same for $\frac{\partial \dot{\mathbf{r}}_{C_A}}{\partial \dot{\theta}_A}$ using the derivative found in Eq. (28). $\dot{\mathbf{r}}_H$ is not a function of $\dot{\theta}_A$:

$$\begin{aligned} \frac{\partial \dot{\mathbf{r}}_{C_A}}{\partial \dot{\theta}_A} &= \frac{\partial}{\partial \dot{\theta}_A} \left(\dot{\mathbf{r}}_H - d \sin \theta_A \dot{\theta}_A \hat{\mathbf{n}}_1 + d \cos \theta_A \dot{\theta}_A \hat{\mathbf{n}}_2 \right) \\ &= \frac{\partial \dot{\mathbf{r}}_H}{\partial \dot{\theta}_A} + d \left(-\sin \theta_A \frac{\partial \dot{\theta}_A}{\partial \dot{\theta}_A} \hat{\mathbf{n}}_1 + \cos \theta_A \frac{\partial \dot{\theta}_A}{\partial \dot{\theta}_A} \hat{\mathbf{n}}_2 \right) \\ &= d(-\sin \theta_A \hat{\mathbf{n}}_1 + \cos \theta_A \hat{\mathbf{n}}_2) \end{aligned} \quad (35)$$

Thus we see that the cancellation-of-dots identity is verified:

$$\frac{\partial \mathbf{r}_{C_A}}{\partial \theta_A} = d(-\sin \theta_A \hat{\mathbf{n}}_1 + \cos \theta_A \hat{\mathbf{n}}_2) = \frac{\partial \dot{\mathbf{r}}_{C_A}}{\partial \dot{\theta}_A} \quad (36)$$

(b.4): Find

Derive the Lagrange equation for θ_A .

Analysis/Conclusions:

We can find the Lagrange equation through the standard form of Lagrange's equation for conservative systems:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0, \quad \forall i \quad (37)$$

Let us identify that in this context there is only one degree of freedom, which is θ_A . Since $\mathbf{q}$ is a minimal set, there is only one generalized coordinate $q_1 = \theta_A$.

This means Eq. (37) rewrites to:

$$\frac{d}{dt} \left(\frac{\partial L_A}{\partial \dot{\theta}_A} \right) - \frac{\partial L_A}{\partial \theta_A} = 0 \quad (38)$$

Then first we find $\frac{\partial L_A}{\partial \theta_A}$ using Eq. (33):

$$\begin{aligned} \frac{\partial L_A}{\partial \theta_A} &= \frac{\partial}{\partial \theta_A} \left(\frac{m_A}{2} X_B^2 \Omega^2 \sin^2 \Omega t + m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A \right. \\ &\quad \left. + \frac{m_A d^2 + J_{CA}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= \frac{\partial}{\partial \theta_A} \left(\frac{m_A}{2} X_B^2 \Omega^2 \sin^2 \Omega t \right) + \frac{\partial}{\partial \theta_A} \left(m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A \right) \\ &\quad + \frac{\partial}{\partial \theta_A} \left(\frac{m_A d^2 + J_{CA}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial \theta_A} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= 0 + m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \frac{\partial}{\partial \theta_A} (\sin \theta_A) + 0 - \frac{k_H}{2} \frac{\partial}{\partial \theta_A} (\theta_A - \theta_0)^2 \\ &= m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \frac{\partial}{\partial \theta_A} (\sin \theta_A) - \frac{k_H}{2} \left(\frac{\partial \theta_A^2}{\partial \theta_A} - 2 \frac{\partial \theta_A}{\partial \theta_A} \theta_0 + \frac{\partial \theta_0^2}{\partial \theta_A} \right) \\ &= m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \cos \theta_A - \frac{k_H}{2} (2\theta_A - 2(1)\theta_0 + 0) \\ &= m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \cos \theta_A - k_H (\theta_A - \theta_0) \end{aligned} \quad (39)$$

Then we find $\frac{\partial L_A}{\partial \dot{\theta}_A}$ using Eq. (33) also:

$$\begin{aligned}
\frac{\partial L_A}{\partial \dot{\theta}_A} &= \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{m_A}{2} X_B^2 \Omega^2 \sin^2 \Omega t + m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A \right. \\
&\quad \left. + \frac{m_A d^2 + J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{m_A}{2} X_B^2 \Omega^2 \sin^2 \Omega t \right) + \frac{\partial}{\partial \dot{\theta}_A} \left(m_A X_B \Omega d \dot{\theta}_A \sin \Omega t \sin \theta_A \right) \\
&\quad + \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{m_A d^2 + J_{C_A}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= 0 + m_A X_B \Omega d \frac{\partial \dot{\theta}_A}{\partial \dot{\theta}_A} \sin \Omega t \sin \theta_A + \frac{m_A d^2 + J_{C_A}}{2} \frac{\partial \dot{\theta}_A^2}{\partial \dot{\theta}_A} - 0 \\
&= m_A X_B \Omega d \sin \Omega t \sin \theta_A + (m_A d^2 + J_{C_A}) \dot{\theta}_A
\end{aligned} \tag{40}$$

Then we find the time derivative $\frac{d}{dt} \left(\frac{\partial L_A}{\partial \dot{\theta}_A} \right)$:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L_A}{\partial \dot{\theta}_A} \right) &= \frac{d}{dt} \left(m_A X_B \Omega d \sin \Omega t \sin \theta_A + (m_A d^2 + J_{C_A}) \dot{\theta}_A \right) \\
&= m_A X_B \Omega d \left(\frac{d}{dt} \sin \Omega t \right) \sin \theta_A \\
&\quad + m_A X_B \Omega d \sin \Omega t \left(\frac{d}{d\theta_A} \sin \theta_A \frac{d\theta_A}{dt} \right) + (m_A d^2 + J_{C_A}) \frac{d\dot{\theta}_A}{dt} \\
&= m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A \\
&\quad + m_A X_B \Omega d \sin \Omega t \cos \theta_A \dot{\theta}_A + (m_A d^2 + J_{C_A}) \ddot{\theta}_A
\end{aligned} \tag{41}$$

With these derivatives found, we can now assemble Lagrange's equation for θ_A :

$$\begin{aligned}
0 &= \frac{d}{dt} \left(\frac{\partial L_A}{\partial \dot{\theta}_A} \right) - \frac{\partial L_A}{\partial \theta_A} \\
&= m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + m_A X_B \Omega d \sin \Omega t \cos \theta_A \dot{\theta}_A + (m_A d^2 + J_{C_A}) \ddot{\theta}_A \\
&\quad - m_A X_B \Omega d \sin \Omega t \cos \theta_A \dot{\theta}_A + k_H (\theta_A - \theta_0) \\
&= m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + (m_A d^2 + J_{C_A}) \ddot{\theta}_A + k_H (\theta_A - \theta_0)
\end{aligned} \tag{42}$$

Then we put it in terms of $\ddot{\theta}_A$:

$$\begin{aligned}
- (m_A d^2 + J_{C_A}) \ddot{\theta}_A &= m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + k_H (\theta_A - \theta_0) \\
\ddot{\theta}_A &= \frac{m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + k_H (\theta_A - \theta_0)}{- (m_A d^2 + J_{C_A})}
\end{aligned} \tag{43}$$

Which makes the Lagrange equation:

$$\ddot{\theta}_A = -\frac{m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}} \quad (44)$$

(b.5): Find

Discuss how this EOM compares to the EOM derived in problem set 3(c) with the bias torque τ_H set to zero.

Analysis/Conclusions:

Recall from problem set 3(c), the EOM derived was found to be:

$$\ddot{\theta}_A = \frac{\tau_H - k_H(\theta_A - \theta_0) - m_A d X_B \Omega^2 \cos \Omega t \sin \theta_A}{m_A d^2 + J_{C_A}} \quad (45)$$

If bias torque τ_H is set to zero, it becomes:

$$\begin{aligned} \ddot{\theta}_A &= \frac{0 - k_H(\theta_A - \theta_0) - m_A d X_B \Omega^2 \cos \Omega t \sin \theta_A}{m_A d^2 + J_{C_A}} \\ \ddot{\theta}_A &= -\frac{m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + k_H(\theta_A - \theta_0)}{m_A d^2 + J_{C_A}} \end{aligned} \quad (46)$$

We see that if we move some terms around, when the bias torque is set to zero, the equation from problem set 3(c) is **identical** to the EOM derived in part (b.4) of this problem set.

This is expected because both are effectively derived from D'Alembert's principle. However, in problem set 3, the equation was derived from a two particle system, whereas in this problem set the problem was derived from the scalar kinetic energy equation and spring potential energy equations. Despite the apparent differences in the methods, the exact same EOM are derived for the array angle θ_A .

(c): Given

Suppose the bus is translationally moving as given in Eq. (22) but not rotating (same as the previous problem). Now, let us add the following motor-and-damper torque to the hinge:

$$\tau_H(\dot{\theta}_A, t) = \tau_{H,a}(t) - c_H \dot{\theta}_A, \quad c_H > 0 \quad (47)$$

(c.1): Find

Derive the force Lagrange equation for θ_A .

Analysis/Conclusions:

We can find the Lagrange equation through the Standard form of Lagrange's equation for non-conservative systems:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_{nc,i}, \quad \forall i \quad (48)$$

Effectively, the only difference here versus the conservative version in Eq. (37) is the inclusion of the generalized force terms coming from non-conservative forces $Q_{nc,i}$.

Just like in part (b), since this system has only one degree of freedom due to the prescribed motion, Eq. (48) can be written as:

$$\frac{d}{dt} \left(\frac{\partial L_A}{\partial \dot{\theta}_A} \right) - \frac{\partial L_A}{\partial \theta_A} = Q_{nc,\theta_A} \quad (49)$$

Where L_A is the same Lagrangian found in part (b). This means all that's left to do is solve for the non-conservative forces Q_{nc,θ_A} .

Using the virtual work of the applied forces in generalized coordinates, as given in class:

$$\delta w = \sum_{j=1}^n \sum_{i=1}^N \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j = \sum_{j=1}^n Q_j \delta q_j \quad (50)$$

where N is the number of particles in the system and each generalized force Q_j is defined as in Eq. (8). However, since there is only one dimension in the defined generalized coordinates, the virtual work of the non-conservative forces simplifies to:

$$\delta w_{nc} = Q_{nc,\theta_A} \delta \theta_A = \tau_H \delta \theta_A \quad (51)$$

Where $\tau_H \delta \theta_A$ is the virtual work of the hinge torque, since a torque does work equal to the torque multiplied by the rotation it acts through, and the bus is not rotating, so the hinge torque acts through exactly the virtual rotation of the array $\delta \theta_A$. This means the non-conservative generalized force is simply the motor-and-damper torque:

$$Q_{nc,\theta_A} = \tau_H(\dot{\theta}_A, t) = \tau_{H,a}(t) - c_H \dot{\theta}_A \quad (52)$$

So, then we can substitute the generalized force into Eq. (49) using the Lagrangian derivative terms already calculated in part (b):

$$\begin{aligned}
Q_{\text{nc},\theta_A} &= \frac{d}{dt} \left(\frac{\partial L_A}{\partial \dot{\theta}_A} \right) - \frac{\partial L_A}{\partial \theta_A} \\
\tau_{H,a}(t) - c_H \dot{\theta}_A &= m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + m_A X_B \Omega d \sin \Omega t \cos \theta_A \dot{\theta}_A \\
&\quad + (m_A d^2 + J_{C_A}) \ddot{\theta}_A - m_A X_B \Omega d \sin \Omega t \cos \theta_A \dot{\theta}_A + k_H (\theta_A - \theta_0) \\
&= m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A + (m_A d^2 + J_{C_A}) \ddot{\theta}_A + k_H (\theta_A - \theta_0)
\end{aligned} \tag{53}$$

Then we once again put it in terms of $\ddot{\theta}_A$:

$$\begin{aligned}
(m_A d^2 + J_{C_A}) \ddot{\theta}_A &= \tau_{H,a}(t) - c_H \dot{\theta}_A - m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A - k_H (\theta_A - \theta_0) \\
\ddot{\theta}_A &= \frac{\tau_{H,a}(t) - c_H \dot{\theta}_A - m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A - k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}}
\end{aligned} \tag{54}$$

Which makes the forced Lagrange equation:

$$\ddot{\theta}_A = \frac{\tau_{H,a}(t) - c_H \dot{\theta}_A - m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A - k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}} \tag{55}$$

(c.2): Find

Consider commanding the motor at H to produce a prescribed array motion $\theta_A^*(t)$ while bus B follows the given translational motion Eq. (22). Derive the required motor torque $\tau_{H,a}(t)$. Identify the contributions associated with array inertia, hinge damping, the spring restoring torque, and prescribed bus acceleration.

Analysis/Conclusions:

To derive the required $\tau_{H,a}(t)$, simply put Eq. (55) in terms of $\tau_{H,a}(t)$:

$$\begin{aligned}
\ddot{\theta}_A &= \frac{\tau_{H,a}(t) - c_H \dot{\theta}_A - m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A - k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}} \\
(m_A d^2 + J_{C_A}) \ddot{\theta}_A &= \tau_{H,a}(t) - c_H \dot{\theta}_A - m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A - k_H (\theta_A - \theta_0) \\
\tau_{H,a}(t) &= (m_A d^2 + J_{C_A}) \ddot{\theta}_A + c_H \dot{\theta}_A \\
&\quad + k_H (\theta_A - \theta_0) + m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A
\end{aligned} \tag{56}$$

Then replace the time-varying $\theta_A(t)$ terms with prescribed $\theta_A^*(t)$ terms to get the equation:

$$\tau_{H,a}(t) = (m_A d^2 + J_{C_A}) \ddot{\theta}_A^* + c_H \dot{\theta}_A^* + k_H (\theta_A^* - \theta_0) + m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A^* \quad (57)$$

From Eq. (57), we can identify the following terms:

$(m_A d^2 + J_{C_A}) \ddot{\theta}_A^*$ is the contributing term for the array inertia because it is the only term that scales with the angular acceleration $\ddot{\theta}_A^*$ and that is by definition inertial torque.

$c_H \dot{\theta}_A^*$ is the contributing term for the hinge damping because it scales directly with angular rate, and damping forces/torques always scale from the first derivative term.

$k_H (\theta_A^* - \theta_0)$ is the contributing term for the spring restoring torque because it is scaling with the displacement of the angle θ_A^* from the reference angle θ_0 . This means, the further the displacement from θ_0 , the stronger the torque will push back towards θ_0 .

$m_A X_B \Omega^2 d \cos \Omega t \sin \theta_A^*$ is the contributing term for the prescribed bus acceleration because it is the first term of $\frac{d}{dt} \left(\frac{\partial L_A}{\partial \dot{\theta}_A} \right)$ found in part (b.4), which came from taking the time derivative of $\sin \Omega t$ in the bus velocity, giving the bus acceleration.

(c.3): Find

Now, consider a scenario where the satellite is thrusting (= generating translational acceleration) with $\mathbf{a}_H = a_0 \hat{\mathbf{n}}_1$ for a constant a_0 , and the motor is off ($\tau_{H,a} = 0$) in Eq. (47).

Let us analyze the stability of the system:

- (i) Determine the equilibrium array angles $\bar{\theta}_A$.
- (ii) Linearize the EOM about the determined equilibrium.
- (iii) Analyze its local stability, identifying the condition for the thrust acceleration magnitude a_0 to ensure asymptotic stability.

Analysis/Conclusions:

First we take the time derivative of the hinge velocity found in Eq. (29) to get the hinge acceleration:

$$\begin{aligned}
 \ddot{\mathbf{r}}_H &= \frac{d\dot{\mathbf{r}}_H}{dt} \\
 &= \frac{d}{dt} (-X_B \Omega \sin \Omega t \hat{\mathbf{n}}_1) \\
 &= -X_B \Omega \frac{d}{dt} \sin \Omega t \hat{\mathbf{n}}_1 \\
 &= -X_B \Omega^2 \cos \Omega t \hat{\mathbf{n}}_1 = a_0 \hat{\mathbf{n}}_1
 \end{aligned} \tag{58}$$

Which means that for the purposes of this problem we can make the substitution:

$$-X_B \Omega^2 \cos \Omega t = a_0 \tag{59}$$

Which can be done in the $\ddot{\theta}_A$ equation from (c.1)::

$$\ddot{\theta}_A = \frac{\tau_{H,a}(t) - c_H \dot{\theta}_A + m_A d a_0 \sin \theta_A - k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}} \tag{60}$$

Then recall that the motor torque is off, so $\tau_{H,a} = 0$:

$$\ddot{\theta}_A = \frac{-c_H \dot{\theta}_A + m_A d a_0 \sin \theta_A - k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}} \tag{61}$$

Then, at equilibrium, the derivative terms of θ_A should be zero, which means:

$$0 = \frac{m_A d a_0 \sin \theta_A - k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}} \tag{62}$$

Then we solve for the equilibrium $\bar{\theta}_A$ as:

$$\begin{aligned}
 (m_A d^2 + J_{C_A}) (0) &= -c_H(0) + m_A d a_0 \sin \bar{\theta}_A - k_H (\bar{\theta}_A - \theta_0) \\
 0 &= m_A d a_0 \sin \bar{\theta}_A - k_H (\bar{\theta}_A - \theta_0) \\
 k_H (\bar{\theta}_A - \theta_0) &= m_A d a_0 \sin \bar{\theta}_A
 \end{aligned} \tag{63}$$

(4.2) Lagrange Multipliers for Systems with Constraints

(d): Given

Suppose the bus B is inertially fixed. Use the following coordinates to describe the configuration of A : $\mathbf{q} = [x_{C_A}, y_{C_A}, \theta_A]^T$. The rigid hinge imposes:

$$\psi_1 = x_{C_A} - d \cos \theta_A = 0, \quad \psi_2 = y_{C_A} - d \sin \theta_A = 0 \quad (64)$$

The torsional spring at H has potential $V_H(\theta_A) = \frac{k_H}{2}(\theta_A - \theta_0)^2$. The system is under the applied force at C_A and the applied torque at H , given as $\mathbf{F}_A = F_{x,A}\hat{\mathbf{n}}_1 + F_{y,A}\hat{\mathbf{n}}_2$ and $\tau_{H,a}$, respectively.

(d.1): Find

Derive the Lagrangian $L(\mathbf{q}, \dot{\mathbf{q}}, t)$.

Analysis/Conclusions:

Recall from problem (b.2) that kinetic energy is given by:

$$T_A = \frac{m_A}{2} \dot{\mathbf{r}}_{C_A} \cdot \dot{\mathbf{r}}_{C_A} + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \quad (65)$$

Since the new coordinate set now also includes Cartesian coordinates on the two planar axes x_{C_A} and y_{C_A} , we can say that:

$$\dot{\mathbf{r}}_{C_A} = \dot{x}_{C_A} \hat{\mathbf{n}}_1 + \dot{y}_{C_A} \hat{\mathbf{n}}_2 \quad (66)$$

Since $\dot{\mathbf{r}}_{C_A}$ is just the derivative of position of the array CoM, and x_{C_A} and y_{C_A} are the positional components aligned with basis vectors $\hat{\mathbf{n}}_1$ and $\hat{\mathbf{n}}_2$ respectively.

This means we can write kinetic energy as:

$$\begin{aligned} T_A &= \frac{m_A}{2} \dot{\mathbf{r}}_{C_A} \cdot \dot{\mathbf{r}}_{C_A} + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \\ &= \frac{m_A}{2} (\dot{x}_{C_A} \hat{\mathbf{n}}_1 + \dot{y}_{C_A} \hat{\mathbf{n}}_2) \cdot (\dot{x}_{C_A} \hat{\mathbf{n}}_1 + \dot{y}_{C_A} \hat{\mathbf{n}}_2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \\ &= \frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \end{aligned} \quad (67)$$

Which puts kinetic energy T_A in terms of generalized coordinates $\mathbf{q}$.

Then finding the Lagrangian $L(\mathbf{q}, \dot{\mathbf{q}}, t)$ can be found by substituting Eq. (67) and the potential energy equation $V_H(\theta_A)$ given by the problem into the following:

$$\begin{aligned} L &= T_A - V_H \\ L &= \frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \end{aligned} \quad (68)$$

(d.2): Find

Derive the Lagrange equations using multipliers $\lambda_{H,1}$ and $\lambda_{H,2}$.

Analysis/Conclusions:

Now that we are modeling constraint forces, we must assume $\mathbf{q}$ is now a non-minimal coordinate set. The standard form of Lagrange's equation for non-conservative systems assuming a non-minimal coordinates set expands upon Eq. (48) (note this form is for holonomic constraints):

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_{nc,i} + \sum_{j=1}^m \lambda_j \frac{\partial \psi_j}{\partial q_i}, \quad i = 1, \dots, n \quad (69)$$

Where i indexes the coordinates of $\mathbf{q}$ and m represents the number of constraints indexed by j .

Since there are now three coordinates in $\mathbf{q}$, there are also three Lagrange equations. Below, they will be derived one at a time.

First for the coordinate $q_1 = x_{C_A}$, the Lagrange equation can be written the following way:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_{C_A}} \right) - \frac{\partial L}{\partial x_{C_A}} = Q_{nc,x_{C_A}} + \lambda_1 \frac{\partial \psi_1}{\partial x_{C_A}} + \lambda_2 \frac{\partial \psi_2}{\partial x_{C_A}} \quad (70)$$

Evaluating $\frac{\partial L}{\partial x_{C_A}}$:

$$\begin{aligned} \frac{\partial L}{\partial x_{C_A}} &= \frac{\partial}{\partial x_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= \frac{\partial}{\partial x_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) \right) + \frac{\partial}{\partial x_{C_A}} \left(\frac{J_{C_A}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial x_{C_A}} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= 0 + 0 - 0 = 0 \end{aligned} \quad (71)$$

Evaluating $\frac{\partial L}{\partial \dot{x}_{C_A}}$:

$$\begin{aligned}
\frac{\partial L}{\partial \dot{x}_{C_A}} &= \frac{\partial}{\partial \dot{x}_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= \frac{\partial}{\partial \dot{x}_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) \right) + \frac{\partial}{\partial \dot{x}_{C_A}} \left(\frac{J_{C_A}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial \dot{x}_{C_A}} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= \frac{m_A}{2} \left(\frac{\partial \dot{x}_{C_A}^2}{\partial \dot{x}_{C_A}} + \frac{\partial \dot{y}_{C_A}^2}{\partial \dot{x}_{C_A}} \right) + 0 - 0 \\
&= \frac{m_A}{2} (2\dot{x}_{C_A} + 0) = m_A \dot{x}_{C_A}
\end{aligned} \tag{72}$$

Evaluating $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_{C_A}} \right)$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_{C_A}} \right) = \frac{d}{dt} (m_A \dot{x}_{C_A}) = m_A \frac{d\dot{x}_{C_A}}{dt} = m_A \ddot{x}_{C_A} \tag{73}$$

Evaluating $\frac{\partial \psi_1}{\partial x_{C_A}}$:

$$\frac{\partial \psi_1}{\partial x_{C_A}} = \frac{\partial}{\partial x_{C_A}} (x_{C_A} - d \cos \theta_A) = \frac{\partial x_{C_A}}{\partial x_{C_A}} - \frac{\partial}{\partial x_{C_A}} d \cos \theta_A = 1 - 0 = 1 \tag{74}$$

Evaluating $\frac{\partial \psi_2}{\partial x_{C_A}}$:

$$\frac{\partial \psi_2}{\partial x_{C_A}} = \frac{\partial}{\partial x_{C_A}} (y_{C_A} - d \sin \theta_A) = \frac{\partial y_{C_A}}{\partial x_{C_A}} - \frac{\partial}{\partial x_{C_A}} d \sin \theta_A = 0 - 0 = 0 \tag{75}$$

Then we can find $Q_{\text{nc}, x_{C_A}}$ from Eq. (8) and the given expression for $\mathbf{F}_A$:

$$\begin{aligned}
Q_{\text{nc}, x_{C_A}} &= \mathbf{F}_A \cdot \frac{\partial \mathbf{r}_{C_A}}{\partial x_{C_A}} \\
&= (F_{x,A} \hat{\mathbf{n}}_1 + F_{y,A} \hat{\mathbf{n}}_2) \cdot \frac{\partial}{\partial x_{C_A}} (x_{C_A} \hat{\mathbf{n}}_1 + y_{C_A} \hat{\mathbf{n}}_2) \\
&= (F_{x,A} \hat{\mathbf{n}}_1 + F_{y,A} \hat{\mathbf{n}}_2) \cdot (1 \hat{\mathbf{n}}_1 + 0 \hat{\mathbf{n}}_2) \\
&= (F_{x,A} \hat{\mathbf{n}}_1 + F_{y,A} \hat{\mathbf{n}}_2) \cdot \hat{\mathbf{n}}_1 = F_{x,A}
\end{aligned} \tag{76}$$

Then plug the above equations back into Eq. (70):

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_{C_A}} \right) - \frac{\partial L}{\partial x_{C_A}} &= Q_{\text{nc}, x_{C_A}} + \lambda_1 \frac{\partial \psi_1}{\partial x_{C_A}} + \lambda_2 \frac{\partial \psi_2}{\partial x_{C_A}} \\
m_A \ddot{x}_{C_A} - 0 &= F_{x,A} + \lambda_1 (1) + \lambda_2 (0) \\
m_A \ddot{x}_{C_A} &= F_{x,A} + \lambda_1 \\
\ddot{x}_{C_A} &= \frac{F_{x,A} + \lambda_1}{m_A}
\end{aligned} \tag{77}$$

Next, for the coordinate $q_2 = y_{C_A}$, the Lagrange equation can be written the following way:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_{C_A}} \right) - \frac{\partial L}{\partial y_{C_A}} = Q_{\text{nc}, y_{C_A}} + \lambda_1 \frac{\partial \psi_1}{\partial y_{C_A}} + \lambda_2 \frac{\partial \psi_2}{\partial y_{C_A}} \quad (78)$$

Evaluating $\frac{\partial L}{\partial y_{C_A}}$:

$$\begin{aligned} \frac{\partial L}{\partial y_{C_A}} &= \frac{\partial}{\partial y_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= \frac{\partial}{\partial y_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) \right) + \frac{\partial}{\partial y_{C_A}} \left(\frac{J_{C_A}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial y_{C_A}} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= 0 + 0 - 0 = 0 \end{aligned} \quad (79)$$

Evaluating $\frac{\partial L}{\partial \dot{y}_{C_A}}$:

$$\begin{aligned} \frac{\partial L}{\partial \dot{y}_{C_A}} &= \frac{\partial}{\partial \dot{y}_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= \frac{\partial}{\partial \dot{y}_{C_A}} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) \right) + \frac{\partial}{\partial \dot{y}_{C_A}} \left(\frac{J_{C_A}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial \dot{y}_{C_A}} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\ &= \frac{m_A}{2} \left(\frac{\partial \dot{x}_{C_A}^2}{\partial \dot{y}_{C_A}} + \frac{\partial \dot{y}_{C_A}^2}{\partial \dot{y}_{C_A}} \right) + 0 - 0 \\ &= \frac{m_A}{2} (0 + 2\dot{y}_{C_A}) = m_A \dot{y}_{C_A} \end{aligned} \quad (80)$$

Evaluating $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_{C_A}} \right)$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_{C_A}} \right) = \frac{d}{dt} (m_A \dot{y}_{C_A}) = m_A \frac{d\dot{y}_{C_A}}{dt} = m_A \ddot{y}_{C_A} \quad (81)$$

Evaluating $\frac{\partial \psi_1}{\partial y_{C_A}}$:

$$\frac{\partial \psi_1}{\partial y_{C_A}} = \frac{\partial}{\partial y_{C_A}} (x_{C_A} - d \cos \theta_A) = \frac{\partial x_{C_A}}{\partial y_{C_A}} - \frac{\partial}{\partial y_{C_A}} d \cos \theta_A = 0 - 0 = 0 \quad (82)$$

Evaluating $\frac{\partial \psi_2}{\partial y_{C_A}}$:

$$\frac{\partial \psi_2}{\partial y_{C_A}} = \frac{\partial}{\partial y_{C_A}} (y_{C_A} - d \sin \theta_A) = \frac{\partial y_{C_A}}{\partial y_{C_A}} - \frac{\partial}{\partial y_{C_A}} d \sin \theta_A = 1 - 0 = 1 \quad (83)$$

Then we can find $Q_{\text{nc}, y_{C_A}}$ from Eq. (8) and the given expression for $\mathbf{F}_A$:

$$\begin{aligned}
Q_{\text{nc}, y_{C_A}} &= \mathbf{F}_A \cdot \frac{\partial \mathbf{r}_{C_A}}{\partial y_{C_A}} \\
&= (F_{x,A} \hat{\mathbf{n}}_1 + F_{y,A} \hat{\mathbf{n}}_2) \cdot \frac{\partial}{\partial y_{C_A}} (x_{C_A} \hat{\mathbf{n}}_1 + y_{C_A} \hat{\mathbf{n}}_2) \\
&= (F_{x,A} \hat{\mathbf{n}}_1 + F_{y,A} \hat{\mathbf{n}}_2) \cdot (0 \hat{\mathbf{n}}_1 + 1 \hat{\mathbf{n}}_2) \\
&= (F_{x,A} \hat{\mathbf{n}}_1 + F_{y,A} \hat{\mathbf{n}}_2) \cdot \hat{\mathbf{n}}_2 = F_{y,A}
\end{aligned} \tag{84}$$

Then plug the above equations back into Eq. (78):

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_{C_A}} \right) - \frac{\partial L}{\partial y_{C_A}} &= Q_{\text{nc}, y_{C_A}} + \lambda_1 \frac{\partial \psi_1}{\partial y_{C_A}} + \lambda_2 \frac{\partial \psi_2}{\partial y_{C_A}} \\
m_A \ddot{y}_{C_A} - 0 &= F_{y,A} + \lambda_1(0) + \lambda_2(1) \\
m_A \ddot{y}_{C_A} &= F_{y,A} + \lambda_2 \\
\ddot{y}_{C_A} &= \frac{F_{y,A} + \lambda_2}{m_A}
\end{aligned} \tag{85}$$

Finally, for the coordinate $q_3 = \theta_A$, the Lagrange equation can be written the following way:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_A} \right) - \frac{\partial L}{\partial \theta_A} = Q_{\text{nc}, \theta_A} + \lambda_1 \frac{\partial \psi_1}{\partial \theta_A} + \lambda_2 \frac{\partial \psi_2}{\partial \theta_A} \tag{86}$$

Evaluating $\frac{\partial L}{\partial \theta_A}$:

$$\begin{aligned}
\frac{\partial L}{\partial \theta_A} &= \frac{\partial}{\partial \theta_A} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= \frac{\partial}{\partial \theta_A} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) \right) + \frac{\partial}{\partial \theta_A} \left(\frac{J_{C_A}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial \theta_A} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= 0 + 0 - \frac{k_H}{2} \left(\frac{\partial \theta_A^2}{\partial \theta_A} - 2 \frac{\partial \theta_A}{\partial \theta_A} \theta_0 + \frac{\partial \theta_0^2}{\partial \theta_A} \right) \\
&= -\frac{k_H}{2} (2\theta_A - 2(1)\theta_0 + 0) = -k_H (\theta_A - \theta_0)
\end{aligned} \tag{87}$$

Evaluating $\frac{\partial L}{\partial \dot{\theta}_A}$:

$$\begin{aligned}
\frac{\partial L}{\partial \dot{\theta}_A} &= \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) + \frac{J_{C_A}}{2} \dot{\theta}_A^2 - \frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{m_A}{2} (\dot{x}_{C_A}^2 + \dot{y}_{C_A}^2) \right) + \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{J_{C_A}}{2} \dot{\theta}_A^2 \right) - \frac{\partial}{\partial \dot{\theta}_A} \left(\frac{k_H}{2} (\theta_A - \theta_0)^2 \right) \\
&= 0 + \frac{J_{C_A}}{2} \frac{\partial \dot{\theta}_A^2}{\partial \dot{\theta}_A} - 0 \\
&= \frac{J_{C_A}}{2} (2\dot{\theta}_A) = J_{C_A} \dot{\theta}_A
\end{aligned} \tag{88}$$

Evaluating $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_A} \right)$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_A} \right) = \frac{d}{dt} (J_{C_A} \dot{\theta}_A) = J_{C_A} \frac{d\dot{\theta}_A}{dt} = J_{C_A} \ddot{\theta}_A \quad (89)$$

Evaluating $\frac{\partial \psi_1}{\partial \theta_A}$:

$$\frac{\partial \psi_1}{\partial \theta_A} = \frac{\partial}{\partial \theta_A} (x_{C_A} - d \cos \theta_A) = \frac{\partial x_{C_A}}{\partial \theta_A} - \frac{\partial}{\partial \theta_A} d \cos \theta_A = 0 - d(-\sin \theta_A) = d \sin \theta_A \quad (90)$$

Evaluating $\frac{\partial \psi_2}{\partial \theta_A}$:

$$\frac{\partial \psi_2}{\partial \theta_A} = \frac{\partial}{\partial \theta_A} (y_{C_A} - d \sin \theta_A) = \frac{\partial y_{C_A}}{\partial \theta_A} - \frac{\partial}{\partial \theta_A} d \sin \theta_A = 0 - d \cos \theta_A = -d \cos \theta_A \quad (91)$$

As discussed in (c.1), the generalized force on an angle coordinate can come directly from a torque, because Eq. (50) defines virtual work as a generalized force times a virtual displacement, and a torque multiplied by a virtual rotation meets the definition of work in rotation. From the problem definition, a single torque $\tau_{H,a}$ is given, so we can say:

$$Q_{nc,\theta_A} = \tau_{H,a} \quad (92)$$

Then plug the above equations back into Eq. (86):

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_A} \right) - \frac{\partial L}{\partial \theta_A} &= Q_{nc,\theta_A} + \lambda_1 \frac{\partial \psi_1}{\partial \theta_A} + \lambda_2 \frac{\partial \psi_2}{\partial \theta_A} \\ J_{C_A} \ddot{\theta}_A - (-k_H (\theta_A - \theta_0)) &= \tau_{H,a} + \lambda_1 (d \sin \theta_A) + \lambda_2 (-d \cos \theta_A) \\ J_{C_A} \ddot{\theta}_A + k_H (\theta_A - \theta_0) &= \tau_{H,a} + \lambda_1 d \sin \theta_A - \lambda_2 d \cos \theta_A \\ J_{C_A} \ddot{\theta}_A &= \tau_{H,a} + \lambda_1 d \sin \theta_A - \lambda_2 d \cos \theta_A - k_H (\theta_A - \theta_0) \\ \ddot{\theta}_A &= \frac{\tau_{H,a} + \lambda_1 d \sin \theta_A - \lambda_2 d \cos \theta_A - k_H (\theta_A - \theta_0)}{J_{C_A}} \end{aligned} \quad (93)$$

Which makes the Lagrange equations:

$$\begin{aligned}
\ddot{x}_{C_A} &= \frac{F_{x,A} + \lambda_{H,1}}{m_A} \\
\ddot{y}_{C_A} &= \frac{F_{y,A} + \lambda_{H,2}}{m_A} \\
\ddot{\theta}_A &= \frac{\tau_{H,a} + \lambda_{H,1}d \sin \theta_A - \lambda_{H,2}d \cos \theta_A - k_H (\theta_A - \theta_0)}{J_{C_A}} \\
\psi_1 &= x_{C_A} - d \cos \theta_A = 0 \\
\psi_2 &= y_{C_A} - d \sin \theta_A = 0
\end{aligned} \tag{94}$$

(d.3): Find

Discuss how the two multipliers are related to the hinge constraint force and its torque about C_A .

Analysis/Conclusions:

We see above in Eq. (94) that the Lagrange multipliers $\lambda_{H,i}$ are treated as straight forces in the directional coordinate acceleration equations, and as forces multiplied by a lever arm in the rotational coordinate one. Since the multiplier $\lambda_{H,1}$ is in the $\ddot{x}_{C_A}$ equation, summed with another force, we see that it is used to represent how the hinge's constraint force pushes and pulls along the $\hat{n}_1$ axis. Likewise, the multiplier $\lambda_{H,2}$ is in the $\ddot{y}_{C_A}$ equation, summed with another force in the same manner, so we see that it represents how the hinge's constraint force pushes and pulls along the $\hat{n}_2$ axis.

$$F_{\text{constraint},x} = \lambda_{H,1}, \quad F_{\text{constraint},y} = \lambda_{H,2} \tag{95}$$

As for how they produce torque, since they are both summed with the external torque $\tau_{H,a}$ and spring restoring torque, they are also part of torque terms scaled by the $d \sin \theta_A$ and $-d \cos \theta_A$ terms respectively. These terms come from the lever arm from C_A . This would have to be the case because whenever a force is applied to the array at H , it would also have to apply a torque with respect to the array CoM. We see this modeled because the entire term is scaled by d which is the distance from H to C_A , and we see that the two Lagrange multipliers end up scaling different terms of a unit circle equation, meaning their torque contributions depend on the current rotation θ_A and scale appropriately on their respective axes.

$$\tau_{\text{constraint}} = \lambda_{H,1}d \sin \theta_A - \lambda_{H,2}d \cos \theta_A \tag{96}$$

(d.4): Find

Discuss why this five-equation formulation (the three Lagrange's equations and the two constraints) is equivalent to the one-coordinate expression in section (4.1), and why the multiplier formulation is useful for hinge-load analysis.

Analysis/Conclusions:

The five-equation formulation with redundant coordinates and the single coordinate expression are equivalent because the end result, being the actual dynamics of the position of C_A with respect to C_B over time, do not change. If you were to numerically integrate both systems of equations over time, both would give you the exact same position of C_A over time. This is because the extra forces that come from the Lagrange multipliers are exactly the forces needed to ensure the hinge H will not move, and the CoM C_A only ever moves on that arc at distance d from H . The one-equation formulation does the same thing by mathematically removing the constraint terms entirely and modifying the remaining equations appropriately, but they both operate on the same principle that the hinge forces perform no work on the system, and they ensure H does not move with respect to C_B .

The reason the redundant-coordinate formulation using Lagrange multipliers is useful for hinge-load analysis is because from these multipliers you can extract functions of constraint forces and torques. While these forces and torques do not perform work from the perspective of the array's dynamics, they could represent external forces and torques for analyzing the static system of the hinge on its own. In other words, the constraint forces could be used to analyze when the external forces pushing and twisting on the hinge would cause the hinge's internal structure to fail.

(d.5): Find

At a given instant, let $\theta_A = \theta_0 = 0$, $\dot{\theta}_A = \omega_0$, $\mathbf{F}_A = \mathbf{0}$, and $\tau_{H,a} = \tau_0$. Do not assume $\ddot{\theta}_A = 0$. Determine $\ddot{\theta}_A$, the hinge constraint force, and its moment about C_A .

Analysis/Conclusions:

From the solution to (d.2) seen in Eq. (94), we can rewrite the constraints as follows:

$$\begin{aligned} x_{C_A} - d \cos \theta_A &= 0 \rightarrow x_{C_A} = d \cos \theta_A \\ y_{C_A} - d \sin \theta_A &= 0 \rightarrow y_{C_A} = d \sin \theta_A \end{aligned} \quad (97)$$

Then we can differentiate these constraints twice to equate them to the $\ddot{x}_{C_A}$ and $\ddot{y}_{C_A}$ equations in Eq. (94):

$$\begin{aligned} \dot{x}_{C_A} &= \frac{dx_{C_A}}{dt} = d \frac{d}{d\theta_A} \cos \theta_A \frac{d\theta_A}{dt} = -d \sin \theta_A \dot{\theta}_A \\ \ddot{x}_{C_A} &= \frac{d\dot{x}_{C_A}}{dt} = -d \left(\frac{d}{d\theta_A} \sin \theta_A \frac{d\theta_A}{dt} \dot{\theta}_A + \sin \theta_A \frac{d\dot{\theta}_A}{dt} \right) = -d \cos \theta_A \dot{\theta}_A^2 - d \sin \theta_A \ddot{\theta}_A \\ \dot{y}_{C_A} &= \frac{dy_{C_A}}{dt} = d \frac{d}{d\theta_A} \sin \theta_A \frac{d\theta_A}{dt} = d \cos \theta_A \dot{\theta}_A \\ \ddot{y}_{C_A} &= \frac{d\dot{y}_{C_A}}{dt} = d \left(\frac{d}{d\theta_A} \cos \theta_A \frac{d\theta_A}{dt} \dot{\theta}_A + \cos \theta_A \frac{d\dot{\theta}_A}{dt} \right) = -d \sin \theta_A \dot{\theta}_A^2 + d \cos \theta_A \ddot{\theta}_A \end{aligned} \quad (98)$$

So then this substitutes back into the $\ddot{x}_{C_A}$ and $\ddot{y}_{C_A}$ equations in Eq. (94) and solves for the Lagrange multipliers as:

$$\begin{aligned} -d \cos \theta_A \dot{\theta}_A^2 - d \sin \theta_A \ddot{\theta}_A &= \frac{F_{x,A} + \lambda_{H,1}}{m_A} \\ -m_A d \cos \theta_A \dot{\theta}_A^2 - m_A d \sin \theta_A \ddot{\theta}_A &= F_{x,A} + \lambda_{H,1} \\ \lambda_{H,1} &= -m_A d \cos \theta_A \dot{\theta}_A^2 - m_A d \sin \theta_A \ddot{\theta}_A - F_{x,A} \\ \lambda_{H,1} &= -m_A d \left(\cos \theta_A \dot{\theta}_A^2 + \sin \theta_A \ddot{\theta}_A \right) - F_{x,A} \\ -d \sin \theta_A \dot{\theta}_A^2 + d \cos \theta_A \ddot{\theta}_A &= \frac{F_{y,A} + \lambda_{H,2}}{m_A} \\ -m_A d \sin \theta_A \dot{\theta}_A^2 + m_A d \cos \theta_A \ddot{\theta}_A &= F_{y,A} + \lambda_{H,2} \\ \lambda_{H,2} &= -m_A d \sin \theta_A \dot{\theta}_A^2 + m_A d \cos \theta_A \ddot{\theta}_A - F_{y,A} \\ \lambda_{H,2} &= m_A d \left(\cos \theta_A \ddot{\theta}_A - \sin \theta_A \dot{\theta}_A^2 \right) - F_{y,A} \end{aligned} \quad (99)$$

Which gives the solution for the Lagrange multipliers, which can then be substituted back into the equation for $\ddot{\theta}_A$:

$$\begin{aligned}
\ddot{\theta}_A &= \frac{\tau_{H,a} + \lambda_{H,1}d \sin \theta_A - \lambda_{H,2}d \cos \theta_A - k_H (\theta_A - \theta_0)}{J_{C_A}} \\
J_{C_A} \ddot{\theta}_A &= \tau_{H,a} + \lambda_{H,1}d \sin \theta_A - \lambda_{H,2}d \cos \theta_A - k_H (\theta_A - \theta_0) \\
&= \tau_{H,a} + \left(-m_A d \left(\cos \theta_A \dot{\theta}_A^2 + \sin \theta_A \ddot{\theta}_A \right) - F_{x,A} \right) d \sin \theta_A \\
&\quad - \left(m_A d \left(\cos \theta_A \ddot{\theta}_A - \sin \theta_A \dot{\theta}_A^2 \right) - F_{y,A} \right) d \cos \theta_A - k_H (\theta_A - \theta_0) \\
&= \tau_{H,a} - m_A d^2 \sin \theta_A \cos \theta_A \dot{\theta}_A^2 - m_A d^2 \sin^2 \theta_A \ddot{\theta}_A - F_{x,A} d \sin \theta_A \\
&\quad - m_A d^2 \cos^2 \theta_A \ddot{\theta}_A + m_A d^2 \sin \theta_A \cos \theta_A \dot{\theta}_A^2 + F_{y,A} d \cos \theta_A - k_H (\theta_A - \theta_0) \\
&= \tau_{H,a} - m_A d^2 (\sin^2 \theta_A + \cos^2 \theta_A) \ddot{\theta}_A - F_{x,A} d \sin \theta_A + F_{y,A} d \cos \theta_A - k_H (\theta_A - \theta_0) \\
&= \tau_{H,a} - m_A d^2 \ddot{\theta}_A - F_{x,A} d \sin \theta_A + F_{y,A} d \cos \theta_A - k_H (\theta_A - \theta_0) \\
(m_A d^2 + J_{C_A}) \ddot{\theta}_A &= \tau_{H,a} - F_{x,A} d \sin \theta_A + F_{y,A} d \cos \theta_A - k_H (\theta_A - \theta_0) \\
\ddot{\theta}_A &= \frac{\tau_{H,a} - F_{x,A} d \sin \theta_A + F_{y,A} d \cos \theta_A - k_H (\theta_A - \theta_0)}{m_A d^2 + J_{C_A}}
\end{aligned} \tag{100}$$

Then substituting in the given quantities for the problem:

$$\begin{aligned}
\ddot{\theta}_A &= \frac{\tau_0 - (0)d \sin(0) + (0)d \cos(0) - k_H (0 - 0)}{m_A d^2 + J_{C_A}} \\
\ddot{\theta}_A &= \frac{\tau_0}{m_A d^2 + J_{C_A}}
\end{aligned} \tag{101}$$

Which gives the value for $\ddot{\theta}_A$.

Then, to obtain the hinge constraint forces and moment about C_A , we can substitute the Lagrange multiplier solutions into Eq. (95) and Eq. (96) respectively:

$$\begin{aligned}
F_{\text{constraint},x} = \lambda_{H,1} &= -m_A d \left(\cos(0)\omega_0^2 + \sin(0)\ddot{\theta}_A \right) - 0 \\
&= -m_A d \left((1)\omega_0^2 + (0)\ddot{\theta}_A \right) \\
&= -m_A d \omega_0^2 \\
F_{\text{constraint},y} = \lambda_{H,2} &= m_A d \left(\cos(0)\ddot{\theta}_A - \sin(0)\omega_0^2 \right) - 0 \\
&= m_A d \left((1)\ddot{\theta}_A - (0)\omega_0^2 \right) \\
&= m_A d \ddot{\theta}_A = \frac{m_A d \tau_0}{m_A d^2 + J_{C_A}} \tag{102}
\end{aligned}$$

$$\begin{aligned}
\tau_{\text{constraint}} &= \lambda_{H,1} d \sin(0) - \lambda_{H,2} d \cos(0) \\
&= \lambda_{H,1} d(0) - \lambda_{H,2} d(1) \\
&= 0 - \left(\frac{m_A d \tau_0}{m_A d^2 + J_{C_A}} \right) d \\
&= -\frac{m_A d^2 \tau_0}{m_A d^2 + J_{C_A}}
\end{aligned}$$

Which gives the hinge constraint force and its moment about C_A :

$$\begin{aligned}
\mathbf{F}_{\text{constraint}} &= -m_A d \omega_0^2 \hat{\mathbf{n}}_1 + \frac{m_A d \tau_0}{m_A d^2 + J_{C_A}} \hat{\mathbf{n}}_2 \\
\tau_{\text{constraint}} &= -\frac{m_A d^2 \tau_0}{m_A d^2 + J_{C_A}} \tag{103}
\end{aligned}$$

(e): Given

Now, suppose the bus B is rotating in the inertial frame (no translational motion). In particular, the bus undergoes a prescribed planar slew $\theta_B(t)$.

(e.1): Find

Express $\mathbf{r}_{C_A}$ and $\dot{\mathbf{r}}_{C_A}$ as functions of θ_A and $\dot{\theta}_A$. Discuss how this changes the contributions to the array angular velocity and CoM velocity. No full EOM derivation is required.

Analysis/Conclusions:

Recall from part (b.1) that the position of the array CoM can be written as the hinge position plus the array direction scaled by d :

$$\mathbf{r}_{C_A} = \mathbf{r}_H + d\hat{\mathbf{e}}_A \quad (104)$$

From (b.1) we define $\mathbf{r}_H$ and $\hat{\mathbf{e}}_A$ as:

$$\mathbf{r}_H = \mathbf{r}_{C_B} + l\hat{\mathbf{n}}_1, \quad \hat{\mathbf{e}}_A = \cos\theta_A\hat{\mathbf{n}}_1 + \sin\theta_A\hat{\mathbf{n}}_2 \quad (105)$$

The position of B is translationally locked, so the value of $\mathbf{r}_{C_B} = \mathbf{r}_{C_B,0}$ since it is constant for all time. However, now that the bus is allowed to rotate, the values of $\mathbf{r}_H$ and $\hat{\mathbf{e}}_A$ will no longer maintain the same alignment with the basis vectors $\hat{\mathbf{n}}_1$ and $\hat{\mathbf{n}}_2$.

Likewise, we define new unit vectors $\hat{\mathbf{b}}_1$ and $\hat{\mathbf{b}}_2$ which stay locked to the orientation of the bus and start aligned with the $\hat{\mathbf{n}}_i$ basis vectors. Similar to how $\hat{\mathbf{e}}_A$ described where C_A was pointing with respect to $\hat{\mathbf{n}}_1$, we can define a vector $\hat{\mathbf{e}}_B$ which now would be the vector pointing from C_B toward the hinge H with respect to $\hat{\mathbf{n}}_1$. This equation takes the same form, but uses the angle θ_B for the bus's rotation:

$$\hat{\mathbf{e}}_B = \cos\theta_B\hat{\mathbf{n}}_1 + \sin\theta_B\hat{\mathbf{n}}_2 \quad (106)$$

Then the orientation of $\hat{\mathbf{e}}_A$ is put in terms of the $\hat{\mathbf{b}}_i$ basis vectors since the hinge is locked to the bus body:

$$\hat{\mathbf{e}}_A = \cos\theta_A\hat{\mathbf{b}}_1 + \sin\theta_A\hat{\mathbf{b}}_2 \quad (107)$$

And then we can define the relationship between $\hat{\mathbf{b}}_i$ and $\hat{\mathbf{n}}_i$ as:

$$\begin{aligned} \hat{\mathbf{b}}_1 &= \cos\theta_B\hat{\mathbf{n}}_1 + \sin\theta_B\hat{\mathbf{n}}_2 = \hat{\mathbf{e}}_B \\ \hat{\mathbf{b}}_2 &= -\sin\theta_B\hat{\mathbf{n}}_1 + \cos\theta_B\hat{\mathbf{n}}_2 \end{aligned} \quad (108)$$

Where $\hat{\mathbf{b}}_2$ is $\hat{\mathbf{b}}_1$ rotated 90° counterclockwise.

Which makes the equation for $\hat{e}_A$:

$$\begin{aligned}
\hat{e}_A &= \cos \theta_A \hat{b}_1 + \sin \theta_A \hat{b}_2 \\
&= \cos \theta_A (\cos \theta_B \hat{n}_1 + \sin \theta_B \hat{n}_2) + \sin \theta_A (-\sin \theta_B \hat{n}_1 + \cos \theta_B \hat{n}_2) \\
&= \cos \theta_A \cos \theta_B \hat{n}_1 + \cos \theta_A \sin \theta_B \hat{n}_2 - \sin \theta_A \sin \theta_B \hat{n}_1 + \sin \theta_A \cos \theta_B \hat{n}_2 \\
&= (\cos \theta_A \cos \theta_B - \sin \theta_A \sin \theta_B) \hat{n}_1 + (\sin \theta_A \cos \theta_B + \cos \theta_A \sin \theta_B) \hat{n}_2 \\
&= \cos(\theta_B + \theta_A) \hat{n}_1 + \sin(\theta_B + \theta_A) \hat{n}_2
\end{aligned} \tag{109}$$

Where the last line uses the angle addition identities $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$ and $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$.

Since the hinge is also locked to the bus body, it now follows $\hat{b}_1$ instead of $\hat{n}_1$:

$$\mathbf{r}_H = \mathbf{r}_{C_B,0} + l \hat{b}_1 = \mathbf{r}_{C_B,0} + l (\cos \theta_B \hat{n}_1 + \sin \theta_B \hat{n}_2) \tag{110}$$

Then substituting $\mathbf{r}_H$ and $\hat{e}_A$ into $\mathbf{r}_{C_A} = \mathbf{r}_H + d \hat{e}_A$:

$$\begin{aligned}
\mathbf{r}_{C_A} &= \mathbf{r}_H + d \hat{e}_A \\
&= \mathbf{r}_{C_B,0} + l (\cos \theta_B \hat{n}_1 + \sin \theta_B \hat{n}_2) + d (\cos(\theta_B + \theta_A) \hat{n}_1 + \sin(\theta_B + \theta_A) \hat{n}_2) \\
&= \mathbf{r}_{C_B,0} + (l \cos \theta_B + d \cos(\theta_B + \theta_A)) \hat{n}_1 + (l \sin \theta_B + d \sin(\theta_B + \theta_A)) \hat{n}_2
\end{aligned} \tag{111}$$

To simplify the derivative, we use the substitution:

$$u = \theta_B + \theta_A, \quad \dot{u} = \dot{\theta}_B + \dot{\theta}_A \tag{112}$$

Then we take the derivative of this expression to get $\dot{\mathbf{r}}_{C_A}$ (note that since $\mathbf{r}_{C_B,0}$ is the constant initial bus position, its derivative is zero):

$$\begin{aligned}
\dot{\mathbf{r}}_{C_A} &= \frac{d\mathbf{r}_{C_A}}{dt} = \frac{d}{dt} \mathbf{r}_{C_B,0} + \frac{d}{dt} (l \cos \theta_B + d \cos u) \hat{n}_1 + \frac{d}{dt} (l \sin \theta_B + d \sin u) \hat{n}_2 \\
&= 0 + \left(l \frac{d}{d\theta_B} \cos \theta_B \frac{d\theta_B}{dt} + d \frac{d}{du} \cos u \frac{du}{dt} \right) \hat{n}_1 + \left(l \frac{d}{d\theta_B} \sin \theta_B \frac{d\theta_B}{dt} + d \frac{d}{du} \sin u \frac{du}{dt} \right) \hat{n}_2 \\
&= \left(-l \sin \theta_B \dot{\theta}_B - d \sin u \dot{u} \right) \hat{n}_1 + \left(l \cos \theta_B \dot{\theta}_B + d \cos u \dot{u} \right) \hat{n}_2 \\
&= \left(-l \sin \theta_B \dot{\theta}_B - d \sin(\theta_B + \theta_A) (\dot{\theta}_B + \dot{\theta}_A) \right) \hat{n}_1 \\
&\quad + \left(l \cos \theta_B \dot{\theta}_B + d \cos(\theta_B + \theta_A) (\dot{\theta}_B + \dot{\theta}_A) \right) \hat{n}_2
\end{aligned} \tag{113}$$

Which gives the expressions for $\mathbf{r}_{C_A}$ and $\dot{\mathbf{r}}_{C_A}$ as:

$$\begin{aligned}\mathbf{r}_{C_A}(\theta_A, t) &= \mathbf{r}_{C_B,0} + (l \cos \theta_B + d \cos(\theta_B + \theta_A)) \hat{\mathbf{n}}_1 \\ &\quad + (l \sin \theta_B + d \sin(\theta_B + \theta_A)) \hat{\mathbf{n}}_2 \\ \dot{\mathbf{r}}_{C_A}(\theta_A, \dot{\theta}_A, t) &= \left(-l \sin \theta_B \dot{\theta}_B - d \sin(\theta_B + \theta_A) (\dot{\theta}_B + \dot{\theta}_A) \right) \hat{\mathbf{n}}_1 \\ &\quad + \left(l \cos \theta_B \dot{\theta}_B + d \cos(\theta_B + \theta_A) (\dot{\theta}_B + \dot{\theta}_A) \right) \hat{\mathbf{n}}_2\end{aligned}\quad (114)$$

Since the motion of θ_B and by extension the slew rate $\dot{\theta}_B$ is prescribed, these are still just functions of θ_A , and $\dot{\theta}_A$.

Compared to the old equations of $\mathbf{r}_{C_A}$ and $\dot{\mathbf{r}}_{C_A}$ found in (b.1) and Eq. (28) before bus rotation was introduced:

$$\begin{aligned}\mathbf{r}_{C_A} &= \mathbf{r}_H + d (\cos \theta_A \hat{\mathbf{n}}_1 + \sin \theta_A \hat{\mathbf{n}}_2) \\ \dot{\mathbf{r}}_{C_A} &= \dot{\mathbf{r}}_H - d \sin \theta_A \dot{\theta}_A \hat{\mathbf{n}}_1 + d \cos \theta_A \dot{\theta}_A \hat{\mathbf{n}}_2\end{aligned}\quad (115)$$

We see two main changes. First, both equations have terms added on that correspond to the rotation of the bus directly. These can be factored out into circle functions that scale to the length l of the bus CoM to the hinge like so:

$$\begin{aligned}\mathbf{r}_{C_A} &: l (\cos \theta_B \hat{\mathbf{n}}_1 + \sin \theta_B \hat{\mathbf{n}}_2) \\ \dot{\mathbf{r}}_{C_A} &: l (-\sin \theta_B \hat{\mathbf{n}}_1 + \cos \theta_B \hat{\mathbf{n}}_2) \dot{\theta}_B\end{aligned}\quad (116)$$

Which happen to be equivalent to the earlier defined body vectors:

$$\begin{aligned}\mathbf{r}_{C_A} &: l \hat{\mathbf{b}}_1 \\ \dot{\mathbf{r}}_{C_A} &: l \hat{\mathbf{b}}_2 \dot{\theta}_B\end{aligned}\quad (117)$$

Which makes sense because this is effectively modeling the position of H to be l distance along $\hat{\mathbf{b}}_1$ and its rate of change to track around that same point tangent to the circular motion of H being on the edge of the bus.

Then, the other component corresponds to the array itself. In both equations, the array terms take the same form as before, but every θ_A is replaced by $\theta_B + \theta_A$:

$$\begin{aligned}\mathbf{r}_{C_A} &: d (\cos(\theta_B + \theta_A) \hat{\mathbf{n}}_1 + \sin(\theta_B + \theta_A) \hat{\mathbf{n}}_2) \\ \dot{\mathbf{r}}_{C_A} &: d (-\sin(\theta_B + \theta_A) \hat{\mathbf{n}}_1 + \cos(\theta_B + \theta_A) \hat{\mathbf{n}}_2) (\dot{\theta}_B + \dot{\theta}_A)\end{aligned}\quad (118)$$

This is because θ_A is now measured relative to the bus, so the array's orientation in the inertial frame is the sum of the bus angle and the relative

array angle. This means the array's angular velocity is no longer just $\dot{\theta}_A$, but instead $(\dot{\theta}_B + \dot{\theta}_A)$. So even if the array is held fixed relative to the bus ($\dot{\theta}_A = 0$), it still rotates at the bus slew rate $\dot{\theta}_B$.